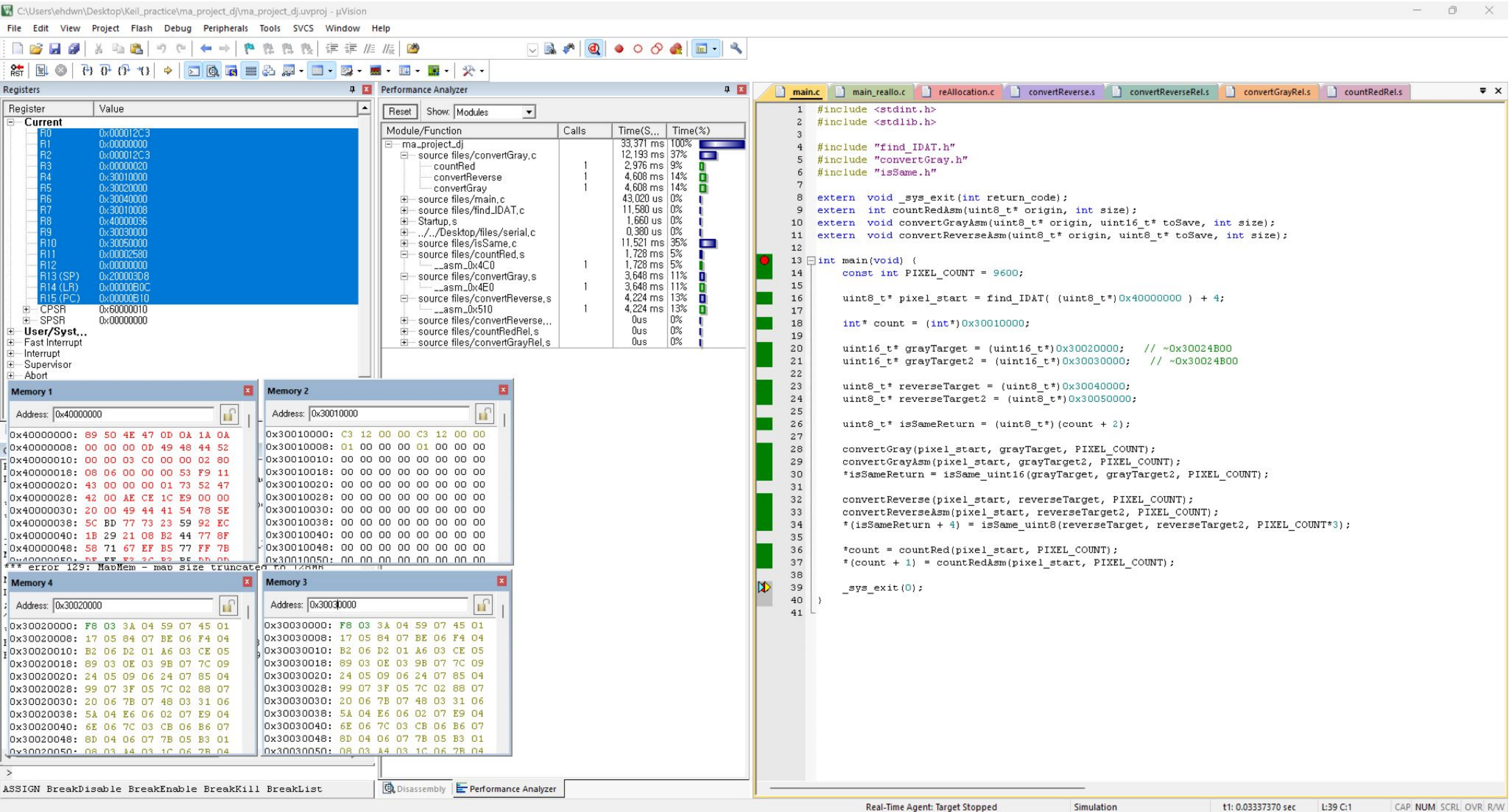
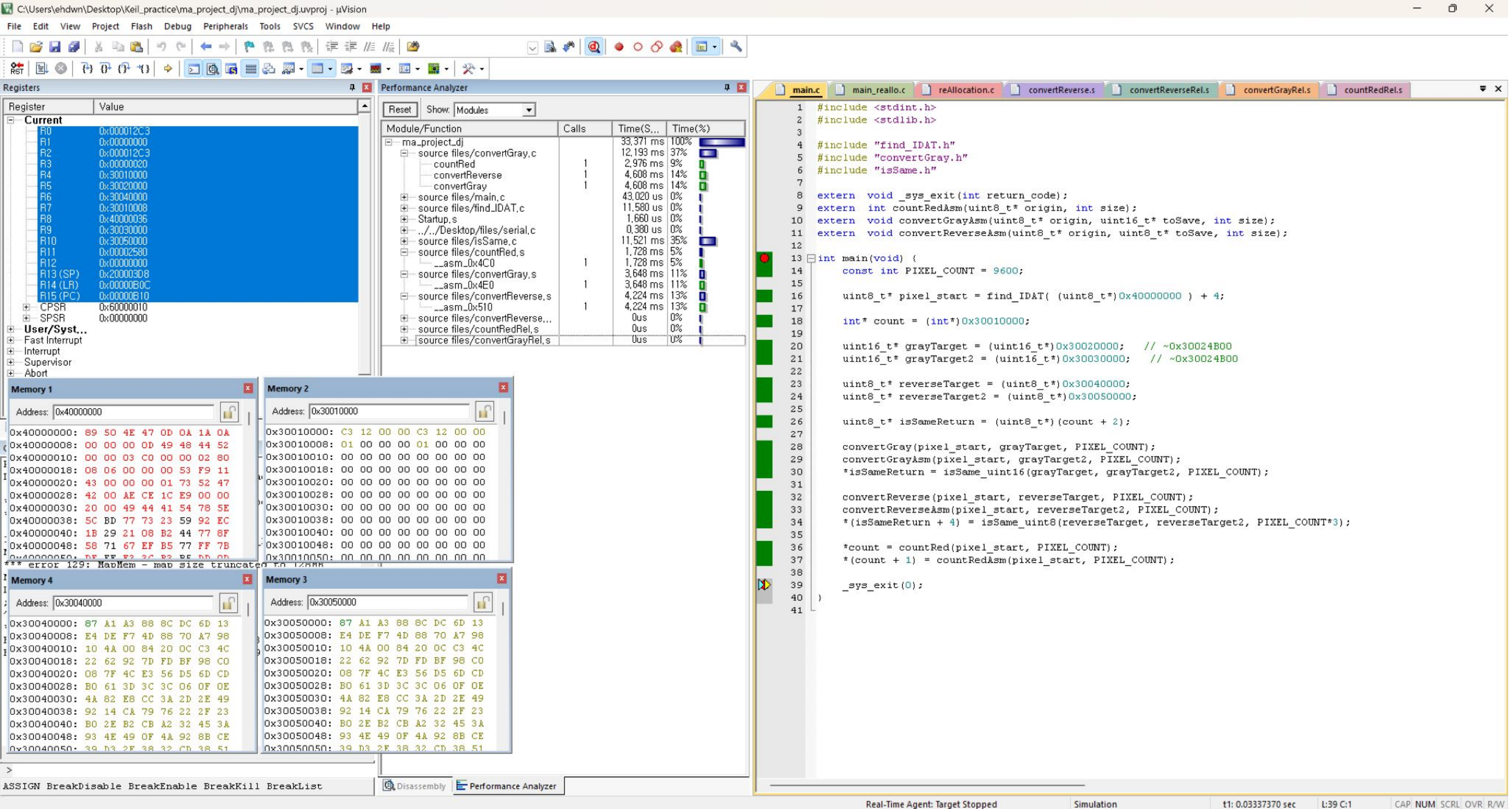
마이크로프로세서 응용 수행 결과

김동준

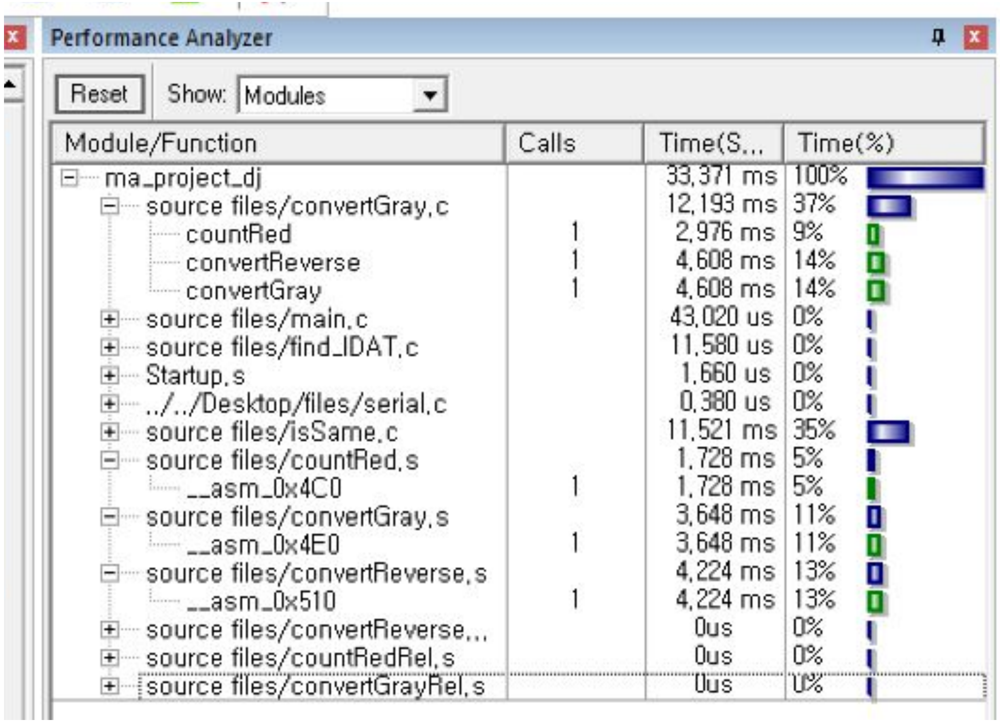
C언어 & ASM 구현 결과 상호 검증(cnt, gray)



C언어 & ASM 구현 결과 상호 검증 (reverse)



수행시간 비교



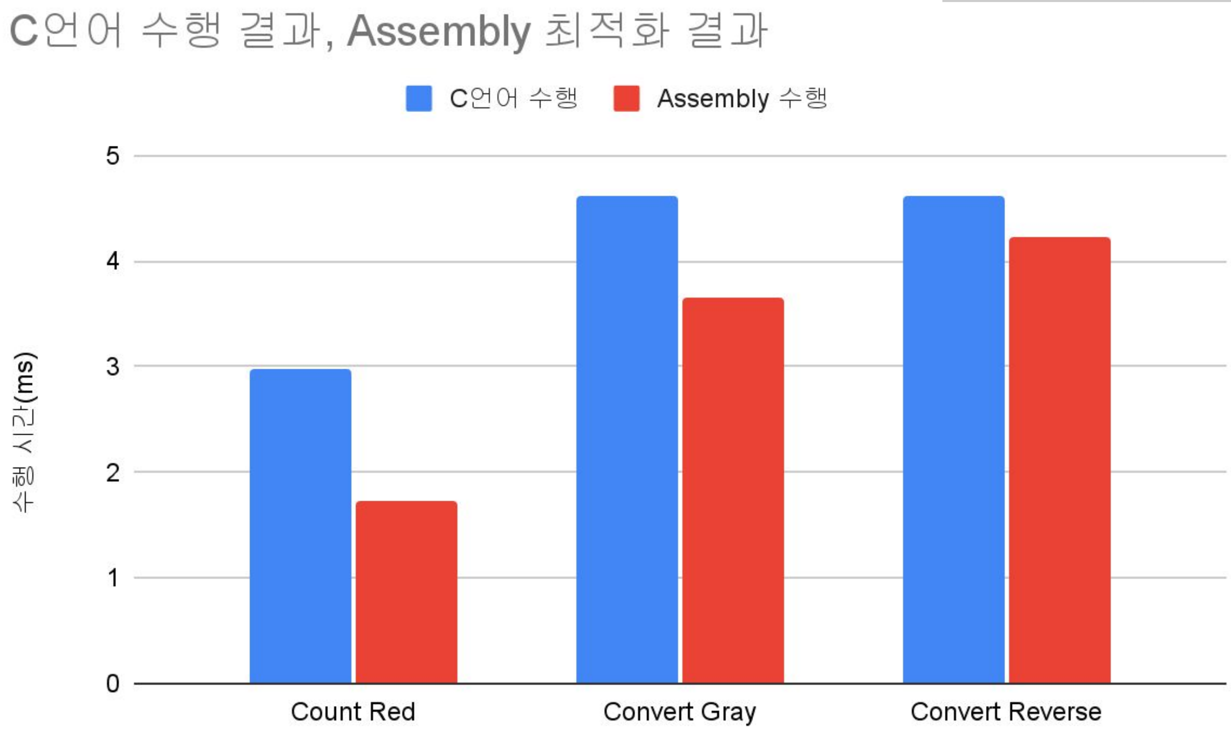
Performance Analyzer

Reset Show: Modules

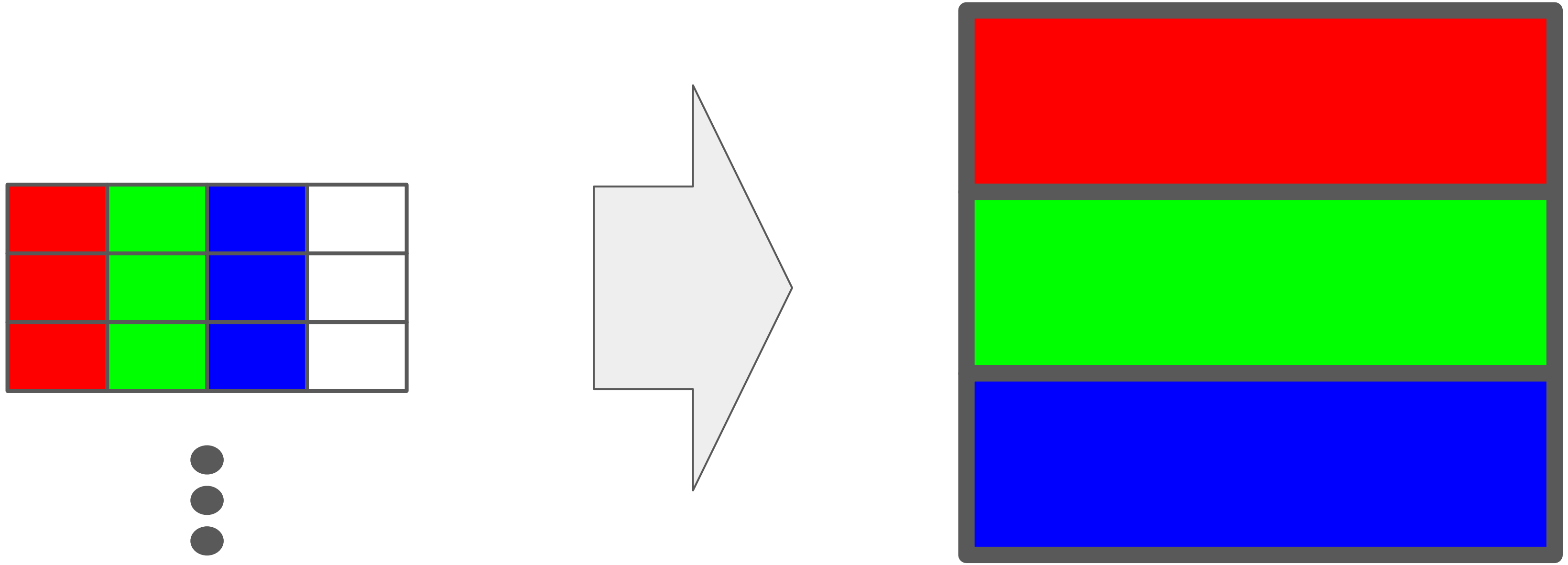
Module/Function	Calls	Time(S...	Time(%)
ma_project_dj		33,371 ms	100%
source files/convertGray.c		12,193 ms	37%
countRed	1	2,976 ms	9%
convertReverse	1	4,608 ms	14%
convertGray	1	4,608 ms	14%
source files/main.c		43,020 us	0%
source files/find_IDAT.c		11,580 us	0%
Startup.s		1,660 us	0%
../../Desktop/files/serial.c		0,380 us	0%
source files/isSame.c		11,521 ms	35%
source files/countRed.s		1,728 ms	5%
__asm_0x4C0	1	1,728 ms	5%
source files/convertGray.s		3,648 ms	11%
__asm_0x4E0	1	3,648 ms	11%
source files/convertReverse.s		4,224 ms	13%
__asm_0x510	1	4,224 ms	13%
source files/convertReverse...		0us	0%
source files/countRedRel.s		0us	0%
source files/convertGrayRel.s		0us	0%

수행시간 비교 시각화

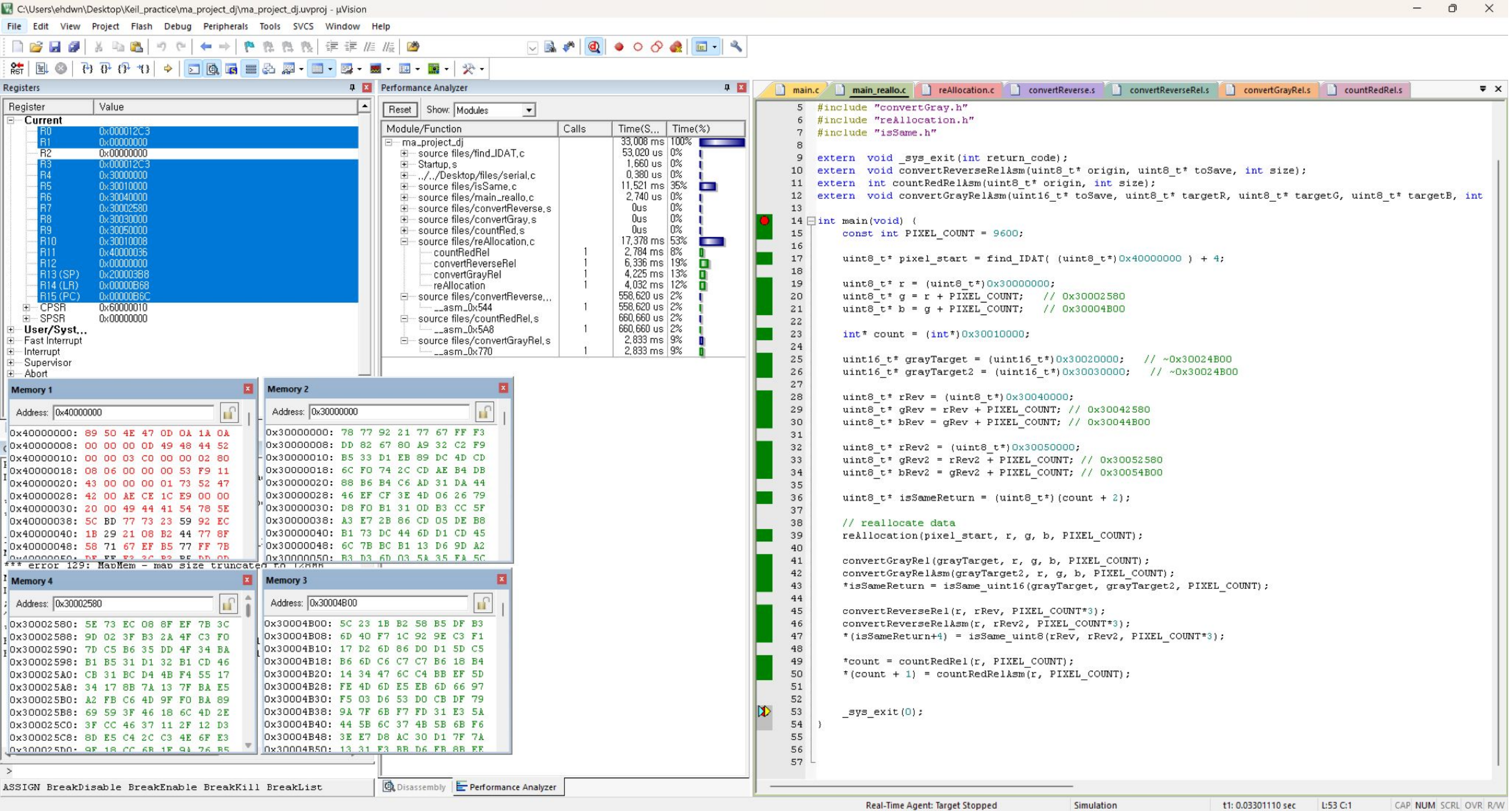
basic		
	C언어 수행 결과	Assembly 최적화 결과
Count Red	2.976	1.728
Convert Gray	4.608	3.648
Convert Reverse	4.608	4.224



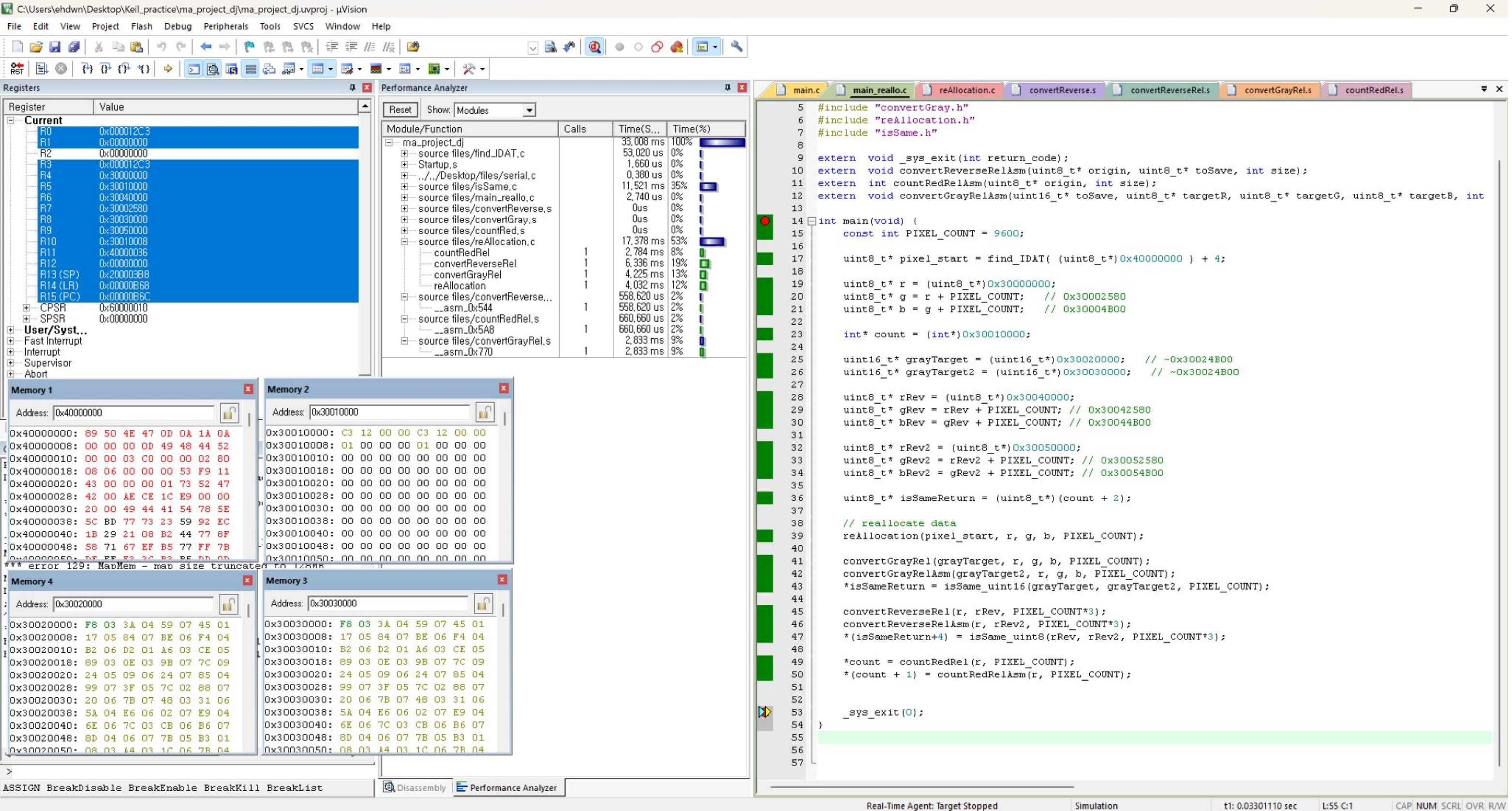
메모리 정렬



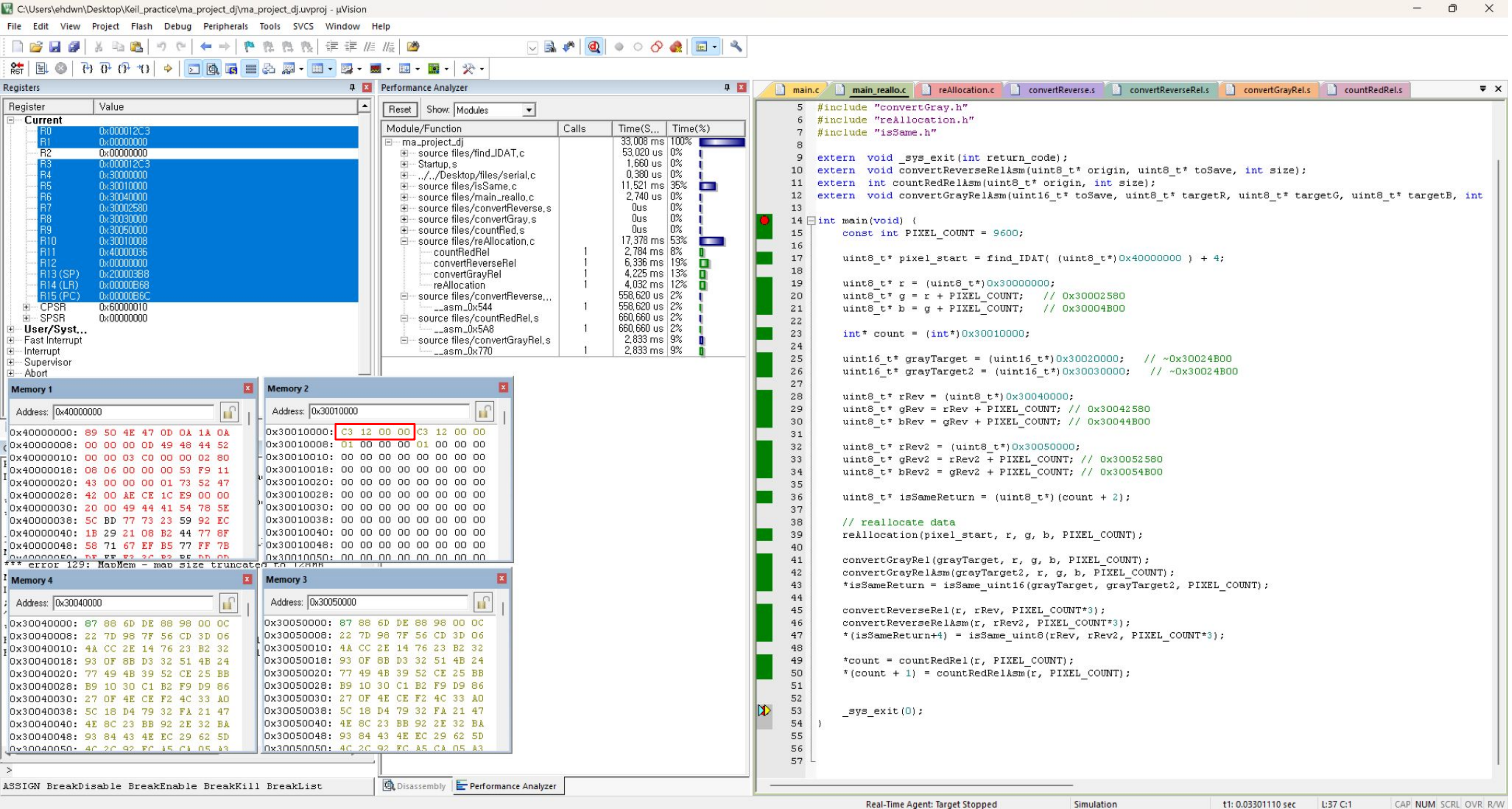
reallocation 구현 결과 검증



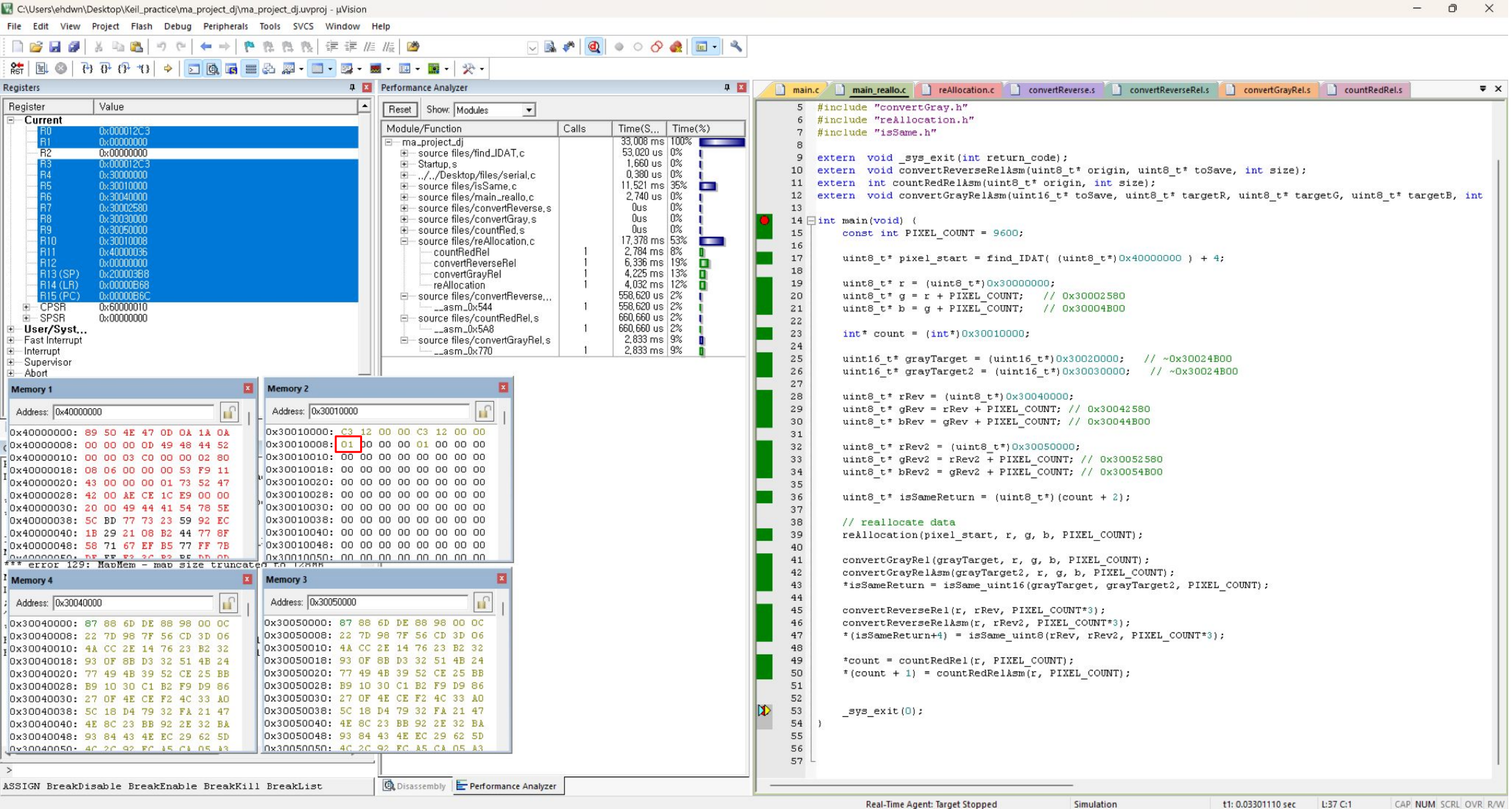
C언어 & ASM 구현 결과 상호 검증(cnt, gray)



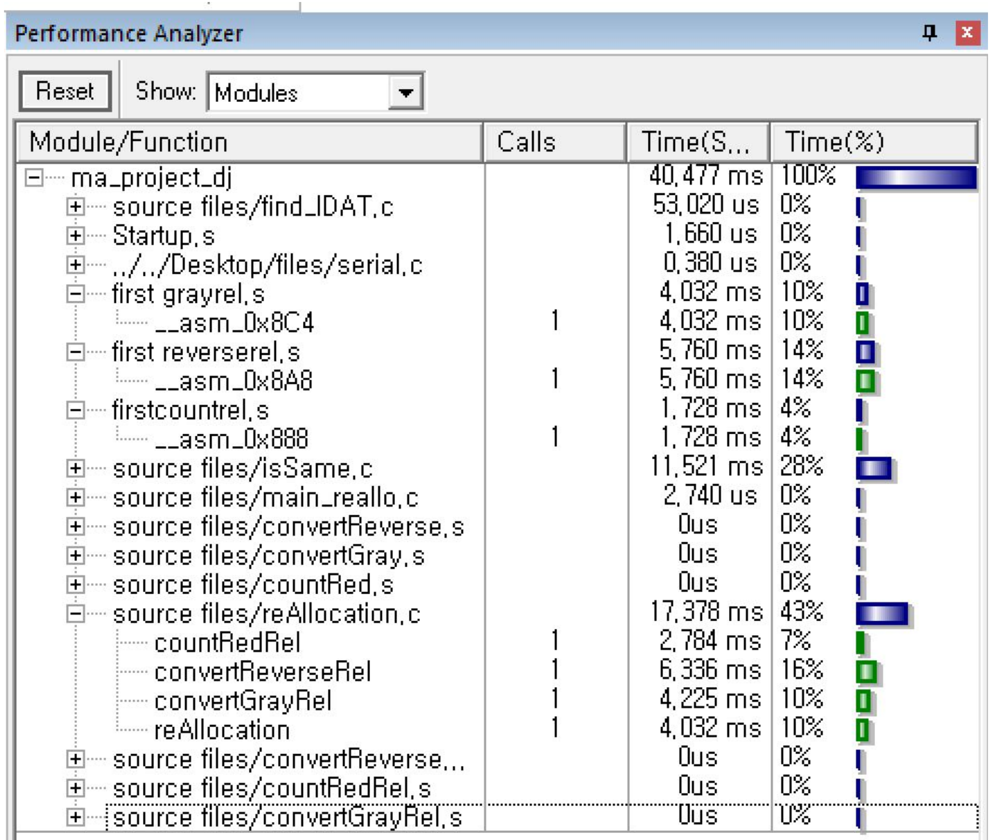
C언어 & ASM 구현 결과 상호 검증 (reverse)



C언어 & ASM 구현 결과 상호 검증 (reverse)



구조 변화 X 초기 수행 속도



Performance Analyzer

Reset Show: Modules

Module/Function	Calls	Time(S...	Time(%)
ma_project_dj		40,477 ms	100%
source files/find_IDAT.c		53,020 us	0%
Startup.s		1,660 us	0%
../../Desktop/files/serial.c		0,380 us	0%
first grayrel.s		4,032 ms	10%
__asm_0x8C4	1	4,032 ms	10%
first reverserel.s		5,760 ms	14%
__asm_0x8A8	1	5,760 ms	14%
firstcountrel.s		1,728 ms	4%
__asm_0x888	1	1,728 ms	4%
source files/isSame.c		11,521 ms	28%
source files/main_reallo.c		2,740 us	0%
source files/convertReverse.s		0us	0%
source files/convertGray.s		0us	0%
source files/countRed.s		0us	0%
source files/reAllocation.c		17,378 ms	43%
countRedRel	1	2,784 ms	7%
convertReverseRel	1	6,336 ms	16%
convertGrayRel	1	4,225 ms	10%
reAllocation	1	4,032 ms	10%
source files/convertReverse...		0us	0%
source files/countRedRel.s		0us	0%
source files/convertGrayRel.s		0us	0%

최적화 결과 수행시간 비교

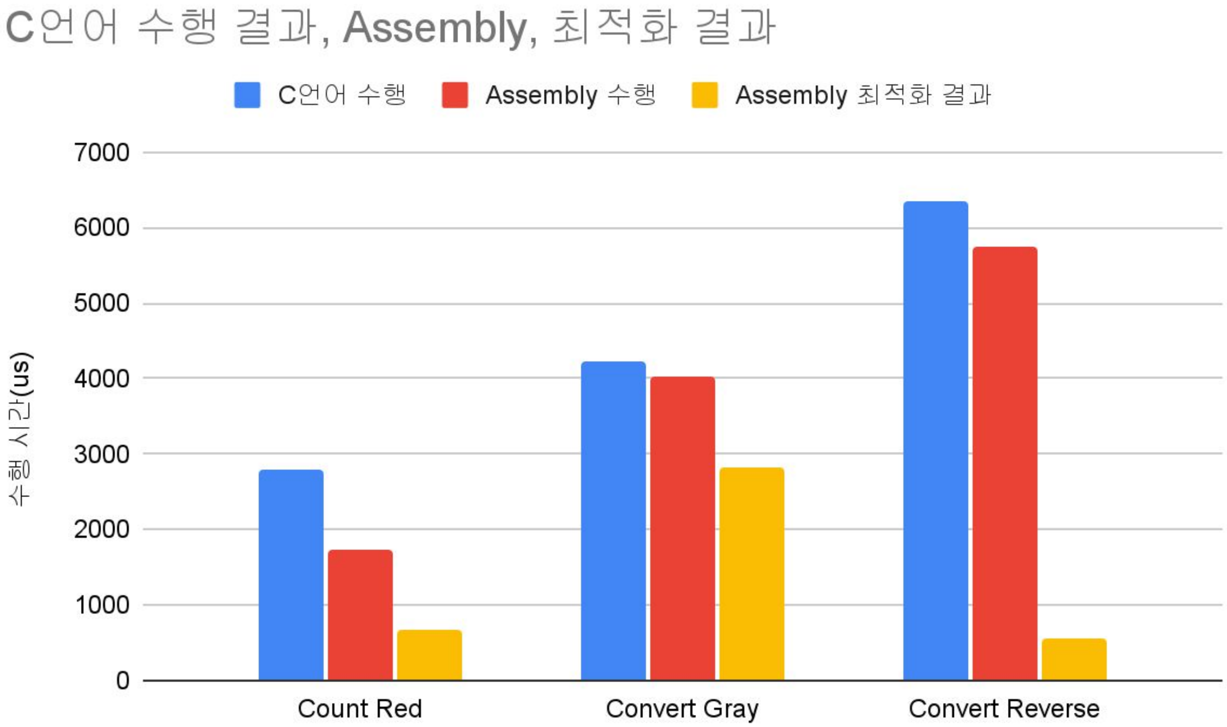
Performance Analyzer

Reset Show: Modules

Module/Function	Calls	Time(S...	Time(%)
ma_project_dj		33,008 ms	100%
source files/find_IDAT.c		53,020 us	0%
Startup.s		1,660 us	0%
../../Desktop/files/serial.c		0,380 us	0%
source files/isSame.c		11,521 ms	35%
source files/main_reallo.c		2,740 us	0%
source files/convertReverse.s		0us	0%
source files/convertGray.s		0us	0%
source files/countRed.s		0us	0%
source files/reAllocation.c		17,378 ms	53%
countRedRel	1	2,784 ms	8%
convertReverseRel	1	6,336 ms	19%
convertGrayRel	1	4,225 ms	13%
reAllocation	1	4,032 ms	12%
source files/convertReverse...		558,620 us	2%
__asm_0x544	1	558,620 us	2%
source files/countRedRel.s		660,660 us	2%
__asm_0x5A8	1	660,660 us	2%
source files/convertGrayRel.s		2,833 ms	9%
__asm_0x770	1	2,833 ms	9%

수행시간 비교 시각화

rellocation			
(us)	C언어 수행	Assembly 수행	Assembly 최적화 결과
Count Red	2784	1728	660.66
Convert Gray	4225	4032	2833
Convert Reverse	6336	5760	558.62



가속 논리

- 잦은 메모리 접근(LDR, STR)은 data bus 점유로 다음 instruction을 느리게 만듦
    - Word, Half word, Byte단위 Load, Store는 모두 같은 사이클 점유
    - LDM, STM을 이용하면 Cycle 절약 가능.
    - 여러 Byte단위(이하 Block 단위)로 Load, Store하여 한 번에 많은 양을 block 단위로 처리
    - Block 단위에 맞지 않는 size의 경우는 일반적인 Byte단위로 처리
- 많은 Loop 횟수는 잦은 Branch로 인해 cycle 손해
    - 고정된 Loop는 횟수 반복문이 아닌 정적인 명령으로 설계(Loop Unrolling)
    - Block 단위 설계를 통한 Loop 횟수 절감 (Loop 횟수 = (Size / Block) + (Size % Block))
- Binary 성질 이용
    - 255 - X = ~X
    - if (X >= 128), X[7] = 1
    - 위의 성질을 이용하여 4 Byte를 하나의 Register에서 연산.

Rverse 연산 설명

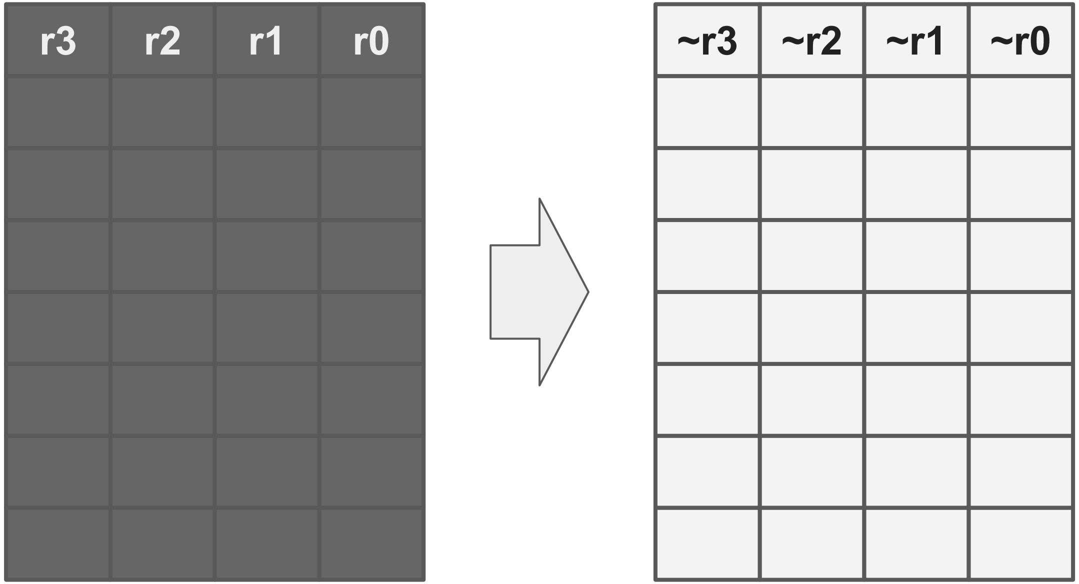
32 Byte 단위 Load,

8번의 word단위 reverse 연산,

32 Byte 단위 Store,

if Size % 32 != 0,

일반적인 Byte 단위 연산



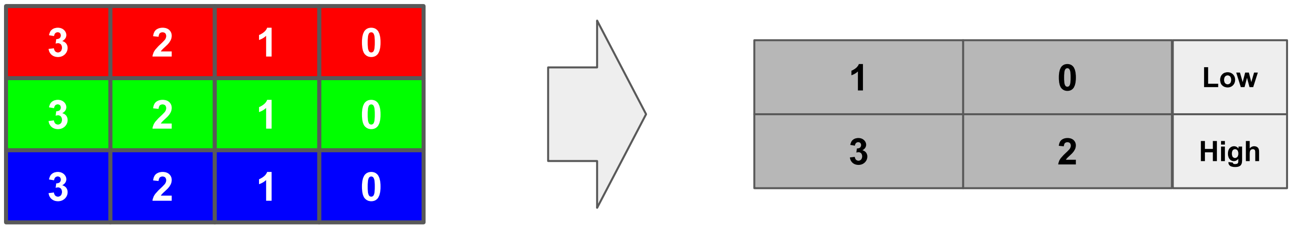
Convert Gray 설명

4 word 단위 R, G, B Load

word에서 R, G, B를 1 Byte씩 가져와 2 Byte Gray pixel 생성,

4개의 r, g, b, 정보를 종합, 2Byte의 Gray pixel 정보를 생성,

4개 픽셀(2 word)를 STM을 이용하여 store



Convert Gray 설명

8 Byte(2 word) 단위 R pixel 정보 Load

Bit mask 사용, 1Byte 단위로 MSB 정보 확인 (TST r0, #128)

Rotate를 이용해서 4pixel 모두 확인

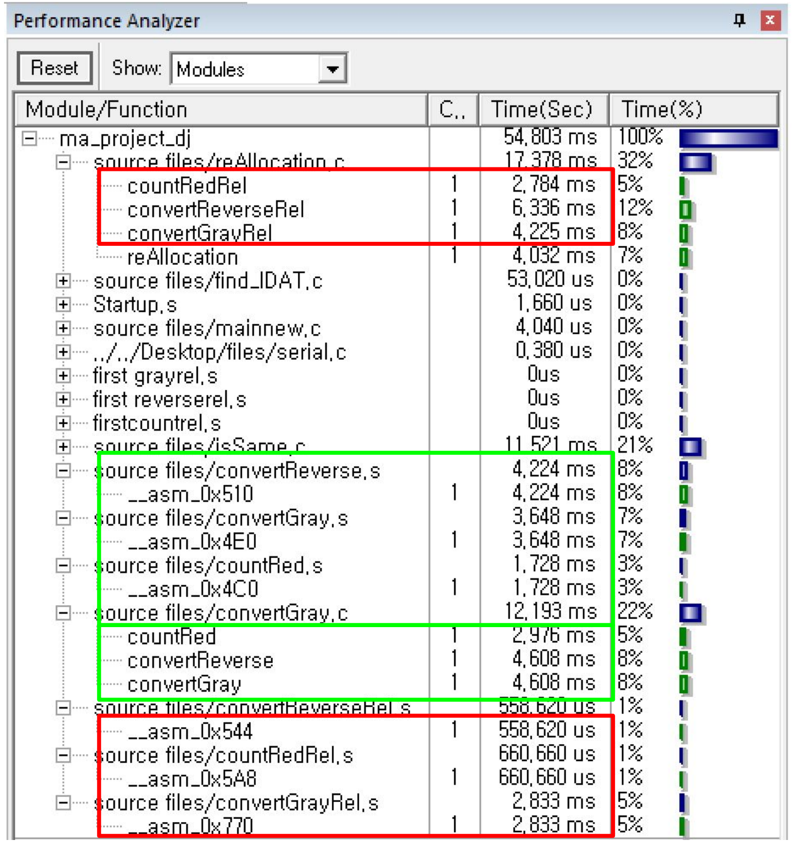
두 word에서 동일한 수행

3	2	1	0
1000_0000	1000_0110	000_1100	1100_0100
0001_0000	0100_0110	000_1100	1110_0100

&

0000_0000	0000_0000	000_0000	1000_0000

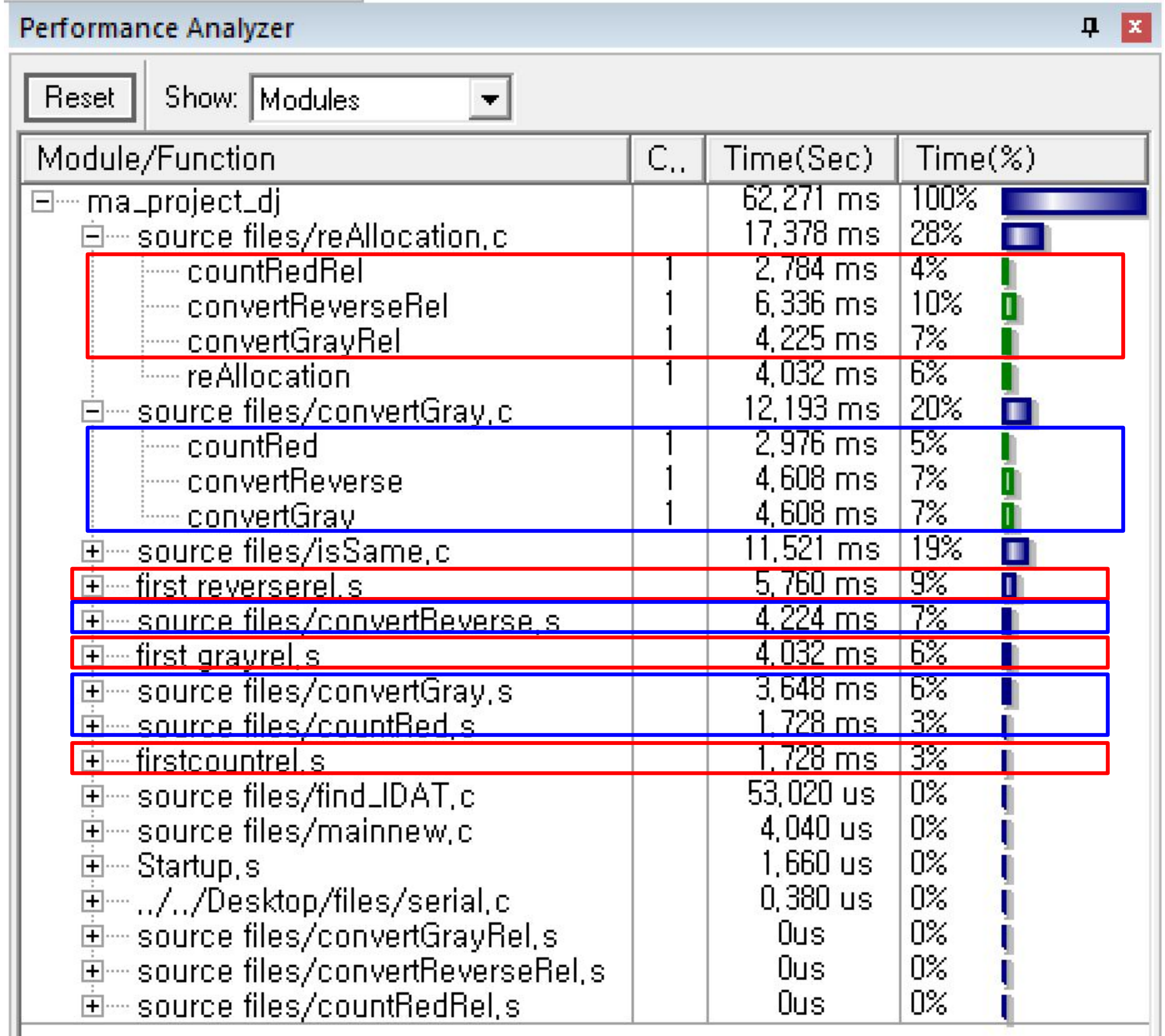
전제 performance



Performance Analyzer

Reset | Show: Modules

Module/Function	C..	Time(Sec)	Time(%)
ma_project_dj		54,803 ms	100%
source files/reAllocation.c		17,378 ms	32%
countRedRel	1	2,784 ms	5%
convertReverseRel	1	6,336 ms	12%
convertGrayRel	1	4,225 ms	8%
reAllocation	1	4,032 ms	7%
source files/find_IDAT.c		53,020 us	0%
Startup.s		1,660 us	0%
source files/mainnew.c		4,040 us	0%
../../Desktop/files/serial.c		0,380 us	0%
first grayrel.s		0us	0%
first reverserel.s		0us	0%
firstcountrel.s		0us	0%
source files/isSame.c		11,521 ms	21%
source files/convertReverse.s		4,224 ms	8%
__asm_0x510	1	4,224 ms	8%
source files/convertGray.s		3,648 ms	7%
__asm_0x4E0	1	3,648 ms	7%
source files/countRed.s		1,728 ms	3%
__asm_0x4C0	1	1,728 ms	3%
source files/convertGray.c		12,193 ms	22%
countRed	1	2,976 ms	5%
convertReverse	1	4,608 ms	8%
convertGray	1	4,608 ms	8%
source files/convertReverseRel.s		558,620 us	1%
__asm_0x544	1	558,620 us	1%
source files/countRedRel.s		660,660 us	1%
__asm_0x5A8	1	660,660 us	1%
source files/convertGrayRel.s		2,833 ms	5%
__asm_0x770	1	2,833 ms	5%



Performance Analyzer

Reset Show: Modules

Module/Function	C..	Time(Sec)	Time(%)
ma_project_dj		62,271 ms	100%
source files/reAllocation.c		17,378 ms	28%
countRedRel	1	2,784 ms	4%
convertReverseRel	1	6,336 ms	10%
convertGrayRel	1	4,225 ms	7%
reAllocation	1	4,032 ms	6%
source files/convertGray.c		12,193 ms	20%
countRed	1	2,976 ms	5%
convertReverse	1	4,608 ms	7%
convertGray	1	4,608 ms	7%
source files/isSame.c		11,521 ms	19%
first reverserel.s		5,760 ms	9%
source files/convertReverse.s		4,224 ms	7%
first grayrel.s		4,032 ms	6%
source files/convertGray.s		3,648 ms	6%
source files/countRed.s		1,728 ms	3%
firstcountrel.s		1,728 ms	3%
source files/find_IDAT.c		53,020 us	0%
source files/mainnew.c		4,040 us	0%
Startup.s		1,660 us	0%
../../Desktop/files/serial.c		0,380 us	0%
source files/convertGrayRel.s		0us	0%
source files/convertReverseRel.s		0us	0%
source files/countRedRel.s		0us	0%